# CO2-AT1 : Chart Construction Test

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## Question 1

# Visualizing a Single Distribution

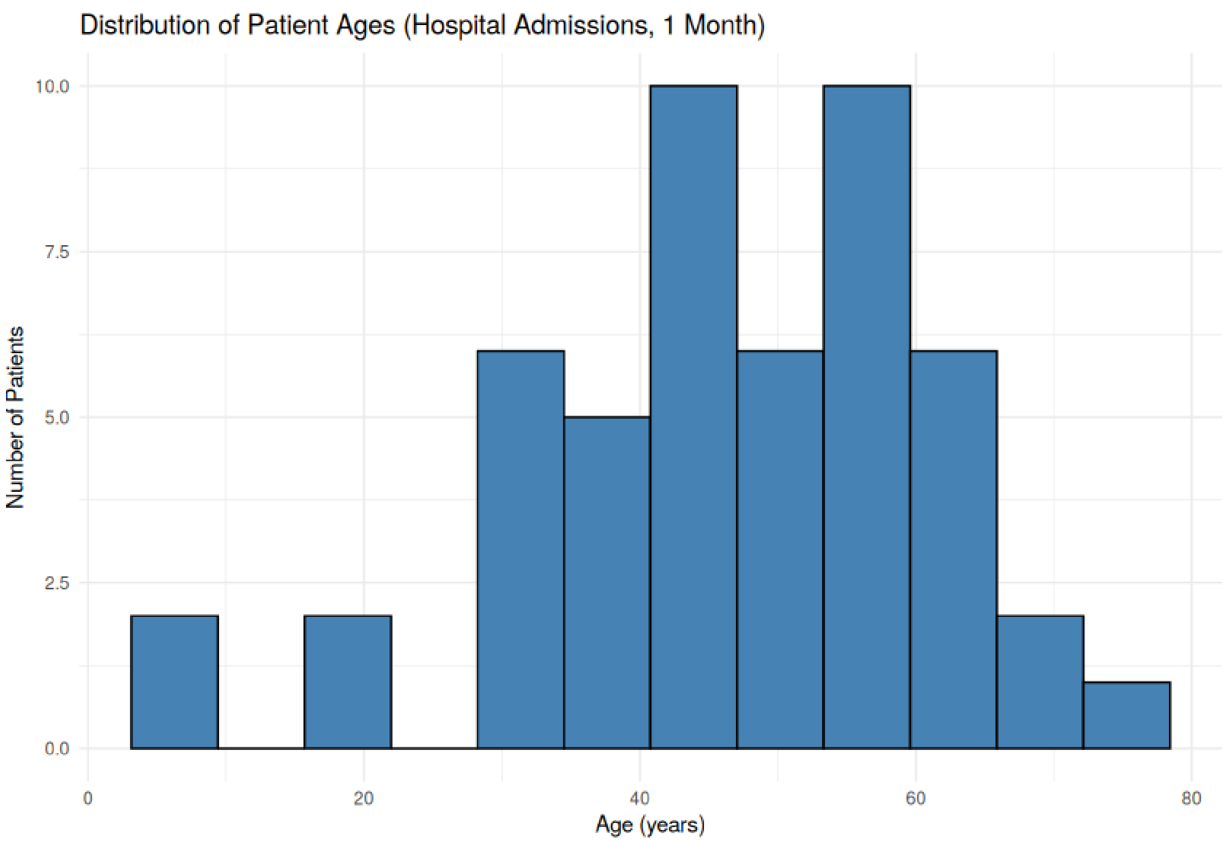
### *Code (R)*

```
library(ggplot2)

# Hardcoded dataset: ages of 50 patients admitted to a hospital in a month
set.seed(1)
patient_ages &lt;- round(rnorm(50, mean = 45, sd = 18))
patient_ages &lt;- pmin(pmax(patient_ages, 1), 95)
df &lt;- data.frame(Age = patient_ages)

# Histogram is the standard choice for visualizing a single continuous
# distribution - it directly shows the shape (spread, skew, modality).
p &lt;- ggplot(df, aes(x = Age)) +
  geom_histogram(bins = 12, fill = "steelblue", color = "black") +
  labs(title = "Distribution of Patient Ages (Hospital Admissions, 1 Month)",
       x = "Age (years)", y = "Number of Patients") +
  theme_minimal()
print(p)
```

### *Output*



## Question 2

# Visualizing Multiple Distributions Simultaneously

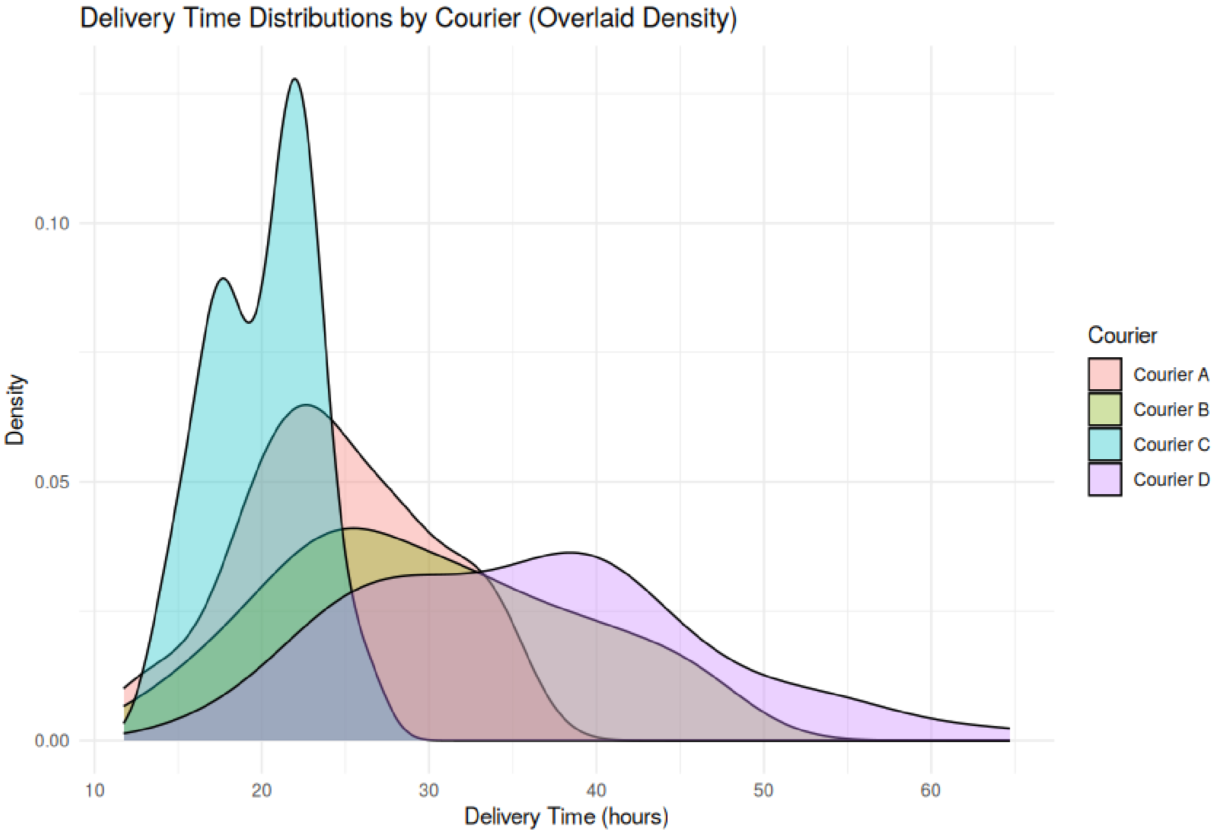
### *Code (R)*

```
library(ggplot2)

# Hardcoded dataset: delivery times (in hours) for 4 different courier companies
set.seed(2)
n &lt;- 60
df &lt;- data.frame(
  DeliveryTime = c(rnorm(n, mean = 24, sd = 5),
                   rnorm(n, mean = 30, sd = 8),
                   rnorm(n, mean = 20, sd = 3),
                   rnorm(n, mean = 36, sd = 10)),
  Courier = rep(c('Courier A', 'Courier B', 'Courier C', 'Courier D'), each = n)
)
df$DeliveryTime &lt;- pmax(df$DeliveryTime, 1)

# Overlaid density plot lets all 4 distributions be compared directly on one
# shared axis, showing differences in center, spread, and shape at a glance.
p &lt;- ggplot(df, aes(x = DeliveryTime, fill = Courier)) +
  geom_density(alpha = 0.35) +
  labs(title = "Delivery Time Distributions by Courier (Overlaid Density)",
       x = "Delivery Time (hours)", y = "Density") +
  theme_minimal()
print(p)
```

### *Output*



## Question 3

# Boxplots for Distribution Comparisons

### *Code (R)*

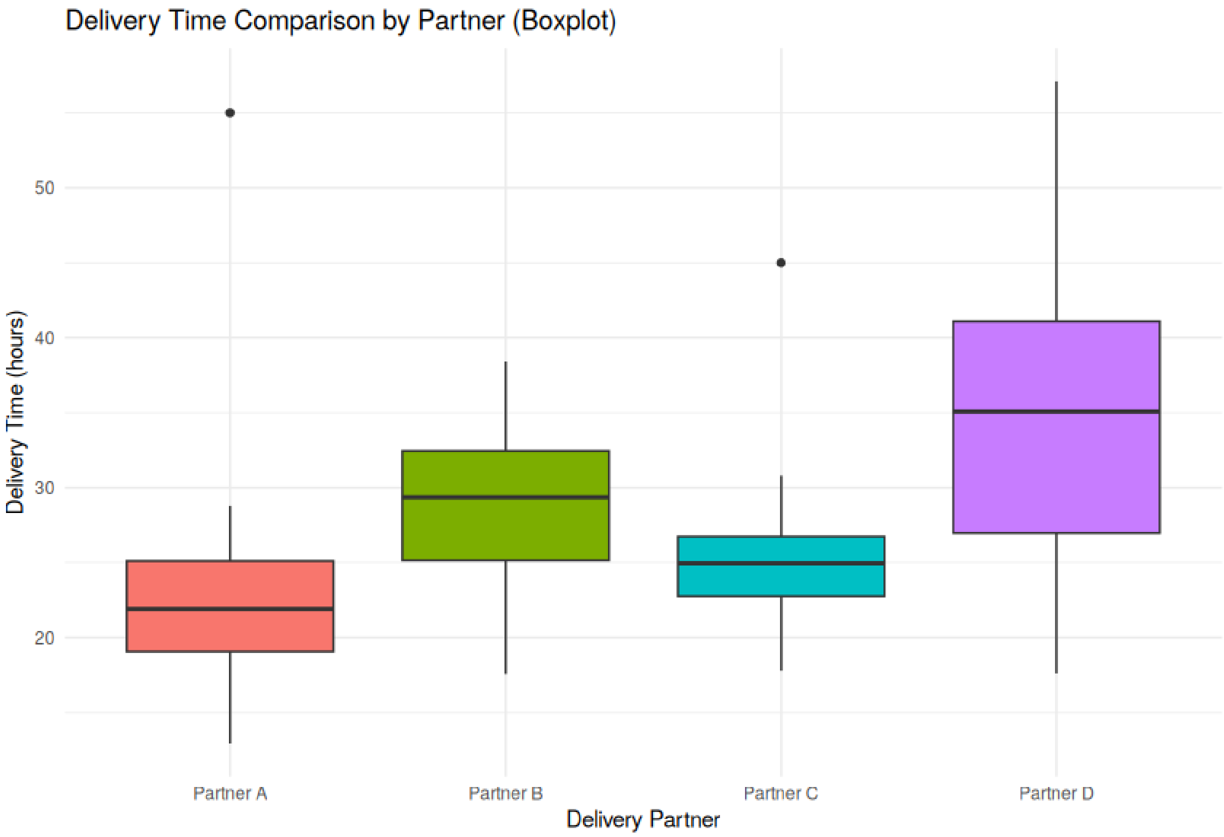
```
library(ggplot2)

# Hardcoded dataset: delivery times (in hours) for 4 different delivery partners
set.seed(3)
n &lt;- 50
df &lt;- data.frame(
  DeliveryTime = c(rnorm(n, mean = 22, sd = 4),
                   rnorm(n, mean = 28, sd = 6),
                   rnorm(n, mean = 25, sd = 3),
                   rnorm(n, mean = 33, sd = 9)),
  Partner = rep(c('Partner A', 'Partner B', 'Partner C', 'Partner D'), each = n)
)
df$DeliveryTime &lt;- pmax(df$DeliveryTime, 1)
# inject a couple of clear outliers to demonstrate outlier identification
df$DeliveryTime[df$Partner == 'Partner A'][1] &lt;- 55
df$DeliveryTime[df$Partner == 'Partner C'][1] &lt;- 45

p &lt;- ggplot(df, aes(x = Partner, y = DeliveryTime, fill = Partner)) +
  geom_boxplot(show.legend = FALSE) +
  labs(title = "Delivery Time Comparison by Partner (Boxplot)",
       x = "Delivery Partner", y = "Delivery Time (hours)") +
  theme_minimal()
print(p)

# Identify outliers per partner using the standard 1.5xIQR rule
for (partner in unique(df$Partner)) {
  sub &lt;- df[df$Partner == partner, ]
  q &lt;- quantile(sub$DeliveryTime, c(0.25, 0.75))
  iqr &lt;- q[2] - q[1]
  lower &lt;- q[1] - 1.5 * iqr
  upper &lt;- q[2] + 1.5 * iqr
  outliers &lt;- sub$DeliveryTime[sub$DeliveryTime &lt; lower | sub$DeliveryTime &gt; upper]
  cat(partner, "- outliers:", if (length(outliers)) round(outliers, 1) else "none", "\n")
}
```

### *Output*



## Question 4

# Amounts

### *Code (R)*

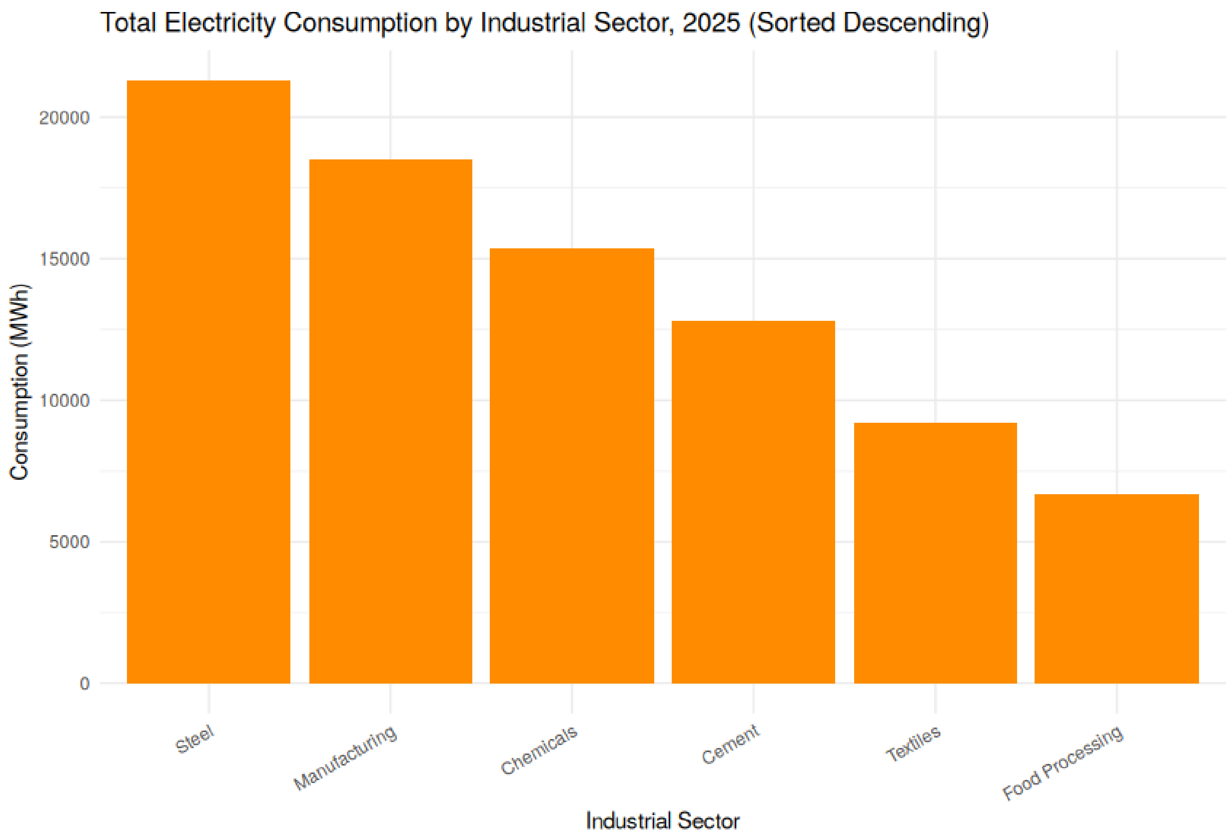
```
library(ggplot2)
library(dplyr)

# Hardcoded dataset: total electricity consumption (MWh) for 6 industrial
# sectors in 2025
df &lt;- data.frame(
  Sector = c('Manufacturing', 'Textiles', 'Chemicals', 'Steel', 'Cement', 'Food Processing'),
  Consumption_MWh = c(18500, 9200, 15400, 21300, 12800, 6700)
)

# Sort descending by consumption before plotting
df &lt;- df %&gt;% arrange(desc(Consumption_MWh))
df$Sector &lt;- factor(df$Sector, levels = df$Sector)

# A bar chart is the standard choice for comparing an "amount" (a single
# numeric quantity) across discrete categories.
p &lt;- ggplot(df, aes(x = Sector, y = Consumption_MWh)) +
  geom_col(fill = "darkorange") +
  labs(title = "Total Electricity Consumption by Industrial Sector, 2025 (Sorted Descending)",
       x = "Industrial Sector", y = "Consumption (MWh)") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 30, hjust = 1))
print(p)
```

### *Output*



## Question 5

# Distributions

### *Code (R)*

```
library(ggplot2)

# Hardcoded dataset: ages of 45 employees in a company
set.seed(5)
employee_ages &lt;- round(rnorm(45, mean = 34, sd = 8))
employee_ages &lt;- pmin(pmax(employee_ages, 21), 60)
df &lt;- data.frame(Age = employee_ages)

# A histogram with a density overlay shows both the discrete frequency
# pattern and a smoothed sense of the overall distribution shape.
p &lt;- ggplot(df, aes(x = Age)) +
  geom_histogram(aes(y = after_stat(density)), bins = 10,
                 fill = "mediumpurple", color = "black", alpha = 0.6) +
  geom_density(color = "black", linewidth = 1) +
  labs(title = "Distribution of Employee Ages (n = 45)",
       x = "Age (years)", y = "Density") +
  theme_minimal()
print(p)
```

### *Output*